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Human Genetic Analysis

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Genetic Analysis – Whole Genome Germline

HG-WHGS-GER

Applications

Method utilizes high-coverage whole genome sequencing (WGS) of genomic DNA to identify genetic variants. Sequencing reads are aligned to a reference genome followed by comprehensive variant calling and annotation. This approach enables detection of single nucleotide variants (SNVs), insertions/deletions (InDels), copy number variations (CNVs), and structural variants relevant to hereditary traits and disease predisposition. Whole genome variant analysis is the most cost-effective and lowest bias method for comprehensive analysis of an individual genome. The method is of choice for discovery og 'genetic disorders' in charecterization of rare inherited disease. The data spans coding and non-coding intergenic regions and is suitable for analysis of copy number variations. A purpose built data annotation system is applied to interrogate the variants and provide up to date annotations, to rank variants by disease phenotypes, accelerating genetic analysis.

Pros

- Comprehensive genome-wide variant detection
- Identifies inherited disease-associated variants
- Suitable for clinical and research studies
- Supports family-based and population genetics
- High sensitivity and accuracy
- Variant annotation with curated datasets

Notes

- Requires high-quality genomic DNA
- Reference map to human reference
- Incidental findings may occur
- Clinical validation recommended for noval variants
- Repeat/long balanced variants may not be called with this application

Input

Genomic DNA (Germline)

DNA Quantity: ≥ 200 ng - 1 µg recommended

High molecular weight DNA analyzed by gel/TapeStation

Sample Types

Blood | Tissue | Buccal Swabs

Platform

Illumina platforms | 30X - 35X | 150 × PE | 100 - 110 GB

Process Map

Sample DNA Extraction > DNA QC > PCR free Library Prep > NGS Data Acquisition > Data QC > Reference Alignment hg19, Grch38 > Variant calling SNP, SNV-indel > CNV analysis > Variant annotation

Standard Deliverables

Download Sample Report

- Sample and Library Quality Control (QC)
- Raw Data Quality Control
- Reference Genome Alignment Summary
- Variant Identification and Summary Statistics
- Zygosity and Coding vs. Non-coding Variant Distribution
- Variant Annotation and Rare Variant Analysis (phenotype-driven, if provided)
- Pathogenic Variant Summary (if identified)
- Copy Number Variation (CNV) Analysis and Variant Effect Prediction

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Exome Sequencing – Standard (Germline) HG-EXOM-STD

Applications

Method utilizes targeted capture of protein-coding regions (exons) followed by high-throughput next-generation sequencing to identify germline genetic variants within clinically and biologically relevant genes. Sequencing reads are aligned to a reference genome, enabling accurate detection of single nucleotide variants (SNVs) and small insertions/deletions (InDels). Exome sequencing provides a cost-effective alternative to whole genome sequencing for studying inherited diseases, population genetics, and gene–phenotype associations.

Pros

- Protein-coding region
- High coverage of genes encoding proteins
- Well-established clinical interpretation frameworks
- Suitable for inherited disease studies as first line investigation of genetic disorders/population analysis

Notes

- Non-coding regions are not captured
- Structural variant detection is limited to short coding genes
- Coverage may vary across GC-rich regions
- Validation recommended for clinically relevant variants

Input

Genomic DNA (Germline)

DNA Quantity: ≥ 100 ng - 1 µg recommended

Sample Types

Blood | Tissue | Buccal Swabs | Cells

Platform

Illumina platforms | 150 × 2 |
100 X Exome, 150 X 2 ~34.9 mb | 8 GB data

Process Map

Sample DNA Extraction > DNA QC > Capture + Library Prep > NGS Data Acquisition > Data QC > Reference Alignment > Variant annotation > Gene Rank by Phenotype > Reporting

Standard Deliverables

Download Sample Report

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary
- Variant Identification and Summary Statistics
- Zygosity and Coding, Variant Annotation and Variant Effect Prediction
- Pathogenic variants(if detected)
- Rare variants by specific phenotypes if described

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Extended Exome Sequencing – Germline

HG-EXOM-EXT

Applications

Method utilizes enhanced target capture of protein-coding regions along with selected regulatory and clinically relevant non-coding regions to comprehensively profile germline genetic variants. Compared to standard exome sequencing, extended exome expands coverage to include untranslated regions (UTRs), splice sites, and disease-associated regulatory elements, improving variant discovery beyond canonical exons. Sequencing reads are aligned to a reference genome followed by variant calling and annotation.

Pros

- Broader coverage target than standard exome
- Improved detection of regulatory variants
- Higher diagnostic yield for rare diseases
- Suitable for unresolved clinical genetic research

Notes

- Intergenic regions out of capture not sequenced
- Structural variant detection is limited
- Coverage variability may occur in GC-rich regions

Input

Genomic DNA (Germline)

DNA Quantity: ≥ 100 ng - 1 µg recommended

Sample Types

Blood | Tissue | Buccal Swabs

Platform

Illumina platforms | 150 × 2
63 mb ~ 14 GB at 100 X

Process Map

Sample DNA Extraction > DNA QC > Capture + Library Prep > NGS Data Acquisition > Data QC & Preprocessing > Reference Alignment > Variant annotation > Gene Rank by Phenotype > Reporting

Standard Deliverables

Download Sample Report

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary
- Variant Identification and Summary Statistics
- Zygosity and Coding, Variant Annotation and Variant Effect Prediction
- Pathogenic variants(if detected)
- Rare variants by specific phenotypes if described
- *Note: For trio 37 MB | 33 GB data/ 11GB/Sample

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Trio Genetic Analysis – Exome (Germline) HG-TRIO-EXO

Applications

Method utilizes exome sequencing of a parent-parent-child trio to identify inherited and de novo germline variants underlying genetic disorders. Protein-coding regions are captured and sequenced, followed by alignment to a reference genome, variant calling, and trio-aware analysis to assess inheritance patterns. This approach improves diagnostic yield by enabling precise filtering of candidate variants based on Mendelian segregation, making it especially effective for rare and undiagnosed conditions. Trio Exome suitable for investigation of rare cariant association, adaptive TDT models and infering transmission of inherited alleles.

Pros

- Higher diagnostic yield than singleton exome
- Accurate detection of idividual variants
- Efficient filtering of benign variants
- Ideal for rare and undiagnosed diseases
- Strong support for clinical interpretation

Notes

- Requires samples from both biological parents + individuals
- Limited to coding and captured regions
- Structural variant detection is limited to indels
- Clinical validation recommended for key findings

Input

Genomic DNA (Germline)

DNA Quantity: 100 ng-1 µg

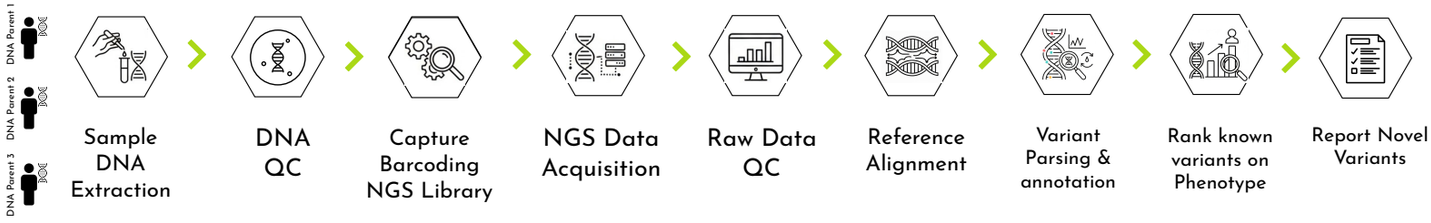
Sample Types

Blood | Tissue | Buccal Swabs

Platform

Illumina platforms | 150 * PE | 100 X Exome | ~35 MB x 3 x 24 GB

Process Map



Standard Deliverables

Download Sample Report

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary- trio
- Variant Summary Trio > Parent 1; Parent 2 > Individual
- Variant annotation and Variant effect prediction
- Novel Variants / Known Variants / Pathogenic Variants (if identified)
- Inheritance pattern analysis - Recessive/ Dominant disorders
- Phenotype based variant ranking (if scoped)

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Research Genome Discovery – Human

HG-RGD-WGS

Applications

Method utilizes comprehensive genome-wide sequencing and analysis to enable hypothesis-driven and discovery-oriented research in human genomics. Long read based whole genome sequencing offers data suitable for detailed profiling of an individual genome. The state of the art method enable4s spanning repeat rich, High GC rich and complex region, without PCR biases to obtain data one genome organization, Structural variations, and genetic variations. The discovery genome pipeline enables germline haplotype phasing, methylation analysis and ranking of variants based on phenotypes providing deeper insights for genetic analysis.

Pros

- Enables unbiased genome-wide discovery
- Suitable for research discovery
- Detects rare and novel genetic/structural variants
- Scalable from small studies to large cohorts
- Supports integrative and population genomics
- Integrates epigenomic signatures and detailed variant annotations

Notes

- For research use only
- Interpretation depends on annotation of variants
- Denovo assembly not standard deliverable

Input

Genomic DNA (Germline)

- DNA Quantity: 200 ng–5 µg
- 5 µg High Molecular Weight DNA recommended

Sample Types

Blood | Tissue | Buccal Swabs | Tumor tissue

Platform

ONT Long-read sequencing platforms

Process Map

Sample DNA Extraction > High Molecular Weight DNA QC > Library Prep/QC > NGS Data Acquisition > Data QC > Reference Alignment > Variant annotation > Reporting

Standard Deliverables

Download Sample Report

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary
- Variant Summary Trio ,SVP/Indel Zygosity, Coding, non-coding variant annotation for both Known and Novel
- Haplotype phasing with long reads
- DNA methylation analysis
- Pathogenic variant summary (if identified)
- CNV analysis and Variant effect prediction
- Structural variants / Repeat > Unaligned read
- *Note: Long read based local analysis or de-novo assembly to be scoped ; Not a standard deliverable

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